

Qatar-1b

R-1954

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_131012 · created 2026-08-06T15:52:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456577.7488 +/- 0.00014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.018
Transit depth (Rp/Rs)^2	1.78 +/- 0.49 [%]
Orbital Inclination (inc)	83.85 +/- 1.72
Airmass coefficient 1 (a1)	1.201 +/- 0.014
Airmass coefficient 2 (a2)	-0.1081 +/- 0.0082
Scatter in the residuals of the lightcurve fit is	1.2 %
Best Comparison Star	None
Optimal Aperture	3.27
Optimal Annulus	11.28
Transit Duration (day)	0.064 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.018 vs 0.1463 (deviation 0.74σ)
Duration fit vs published	0.064 d vs 0.0692 d (deviation 0.43σ)
Mid-time O–C	-0.3 min
Depth SNR	7.4
Residual scatter	1.2 %

Light curve

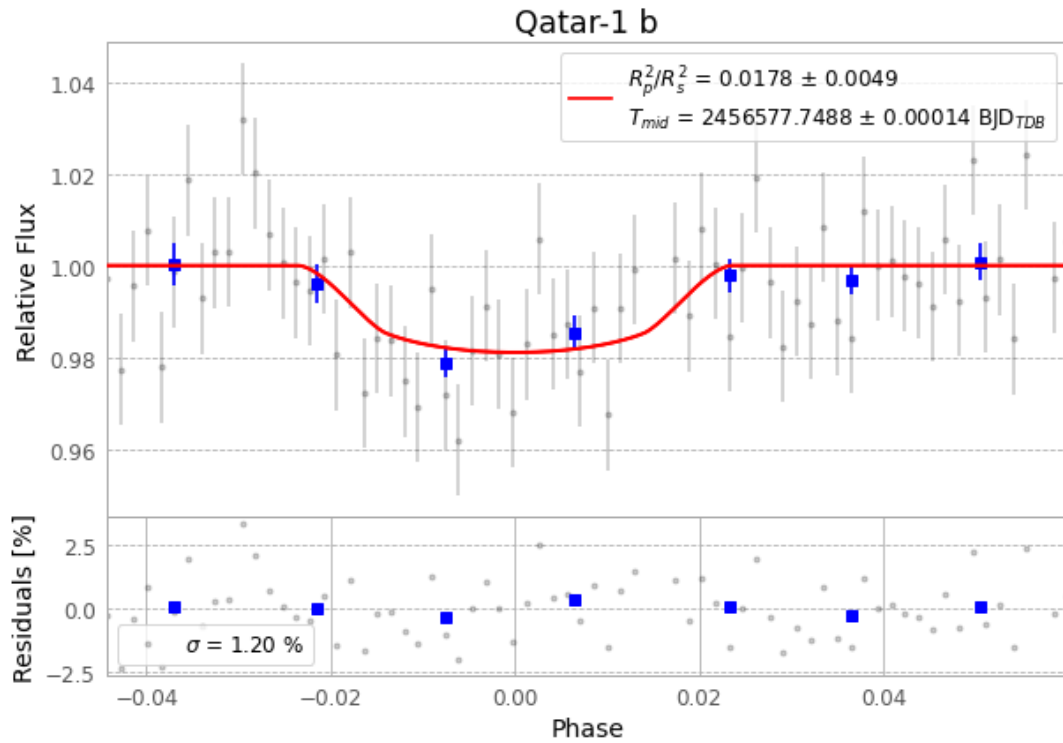


Figure 1 — FinalLightCurve_Qatar-1 b_2013-10-11.png (SHA-256 d2aa56d9be7ff09...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 294], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 45d777748d3f2706...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

72 FITS frames (47 MB), Qatar-1131012042412.FITS → Qatar-1131012075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-131012.log (35aaf27524b9...), FinalLightCurve_Qatar-1 b_2013-10-11.png (d2aa56d9be7f...), FinalLightCurve_Qatar-1 b_2013-10-11.csv (c8ad98f0d9fb...)

Compute resources

Wall time 122.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:52Z.